



Indian Science, Technology and Engineering facilities Map (I-STEM)

Linking Researchers and Resources
Launched by the Honourable Prime Minister of India

Register and Become a member of the Scientific World through the I-STEM National Portal

www.istem.gov.in/supplier/home
<https://www.istem.gov.in/services/home>



Supplier and Service Provider Map for Scientific Community

Benefits to Companies Joining/Registering with the Portal

- ✓ No Registration Fee
- ✓ Access to the I-STEM Community (25K users from Govt. and Pvt. Institutions and National R&D Labs, PSEs, and industry)
- ✓ Access to Webinars/Seminars/Conferences and Events in which to advertise your Company
- ✓ Get Alerts when Equipment is Down/Under Repair (as marked by the custodian institution)
- ✓ Join the I-STEM Comprehensive AMC program

Benefits to the R&D community

- ✓ Users-friendly portal to provide first-hand information about R&D Services and Supplies
- ✓ Easy to contact Service Providers and Suppliers of Accessories
- ✓ Directs you to help in maintaining lab tools efficiently and having their life get extended
- ✓ Opportunity to network across the country and to seek, receive, and provide technical support

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travel adaptors and wireless charging devices, and it is in these markets that GaN has witnessed the highest inflection. Further, the year 2019 saw GaN chargers taking off in a big way, which has further contributed to its high market share.

The power semiconductor market is 44 billion dollars. It is estimated that with SiC and GaN ramping up, they could form ten percent of the market by 2024.

There is a steady growth in their presence in many relatively new market applications as well, like electric vehicles (EVs) and hybrid electric vehicles (HEVs). Even within these segments, they are gaining traction for further sub-segments like charging infrastructure, and energy storage devices.

WBG semiconductor landscape in India

The SiC and GaN semiconductor industry in India has seen a positive growth trend in the past couple of years. Many companies have shown their interest in investing in projects related to WBG semiconductor produc-

tion in India, mainly due to factors like cost efficiency and talent acquisition.

Sunil Kaul, senior advisor at Visicon Power, says, "We believe the overall power electronics market is very relevant for the Indian conditions, and we are looking at doing some work out of India. We are already in discussion with some large OEMs in India about identifying some immediate applications like silicon carbide and hopefully, as time goes by, we will collaborate with them. We are very excited about that."

The Indian academia, along with the various Central government schemes and policies that aim to boost chip manufacturing in India, has paved the way for various large semiconductor manufacturing companies to look towards India as a potentially growing market.

"There is a major project on GaN cleared for the Indian Institute of Science (IISc), Bengaluru, and it is something which we hope to see as a real project on the ground. It has positioned itself as an incubator foundry for GaN. If it is planned right, it can

be very effective for creating the GaN industry in India. In addition to that, several people have applied for a gallium nitride fab, so we are watching that space and are hoping to see some of these projects turn into reality. So, the (GaN) ecosystem is no longer zero," explains Dr Ashwini Aggarwal.

Each power electronics market, whether conventional or new, is evolving to accommodate the use of WBG semiconductors. Although we might not see them being used as the mainstream chips in the manufacturing industry, with initiatives like Visicon's, along with the Central government's consistent efforts to incorporate the use of WBG chips in products, we can hope to see a new revolution coming soon, which will be helmed and accelerated by the WBG semiconductor chip manufacturing industry. **EFY**

